

No. of Functional Features Included in the Solution

Date	27 September 2026
Team ID	SWTID-2026-3342
Project Name	AI BlogNest API
Maximum Marks	5 Marks

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status
1	User Registration	Create an account with name/email/password	authController.js / userModel.js	Done
2	User Login (JWT)	Authenticate credentials and issue a JWT access token	authController.js	Done
3	Role & Auth Middleware	Verify the JWT and enforce Admin/Editor/Author/Reader permissions on protected routes	middleware/	Done
4	Blog CRUD	Create, read, update and delete blog posts; new posts default to draft status	blogController.js / blogModel.js	Done
5	Categories & Tags	Organize posts with a category and an array of tags for filtering/discovery	blogController.js	Done
6	Comment Management	Readers add comments (default status "pending"); Editors/Admins moderate them	blogController.js / commentModel.js	Done
7	AI Blog Generator	Generates a structured post (title + content) from a topic and category via Google Gemini (e.g. "The importance of artificial intelligence in education" → a full guide-style post)	aiController.js / geminiService.js	Done
8	AI Content Summarizer	Condenses existing blog content into a concise summary via Google Gemini	aiController.js / geminiService.js	Done

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Registration, Login, Role/Auth Middleware, Blog CRUD, Categories & Tags, Comment Management)
Additional / Nice-to-Have Features	2 (AI Blog Generator, AI Content Summarizer)
Features Tested & Verified	8 of 8, verified via Postman, 22–27 September 2026

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	0	None – headless REST API, consumed via API clients by design (a <code>Client/</code> folder is reserved for a future UI)
2	Backend / Logic	6	Auth, Blog and AI controllers, plus role middleware and comment moderation logic
3	Database / Storage	3	userModel.js, blogModel.js and commentModel.js Mongoose models
4	API / Integration	1	Google Gemini AI (blog generation & summarization)
5	Security / Authentication	2	JWT authentication, role-based access control